

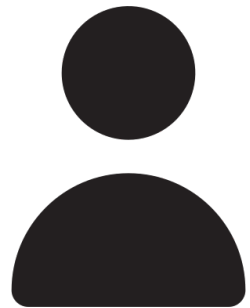


FvKG[json]: Federated Virtual Knowledge Graph over JSON-based Web APIs

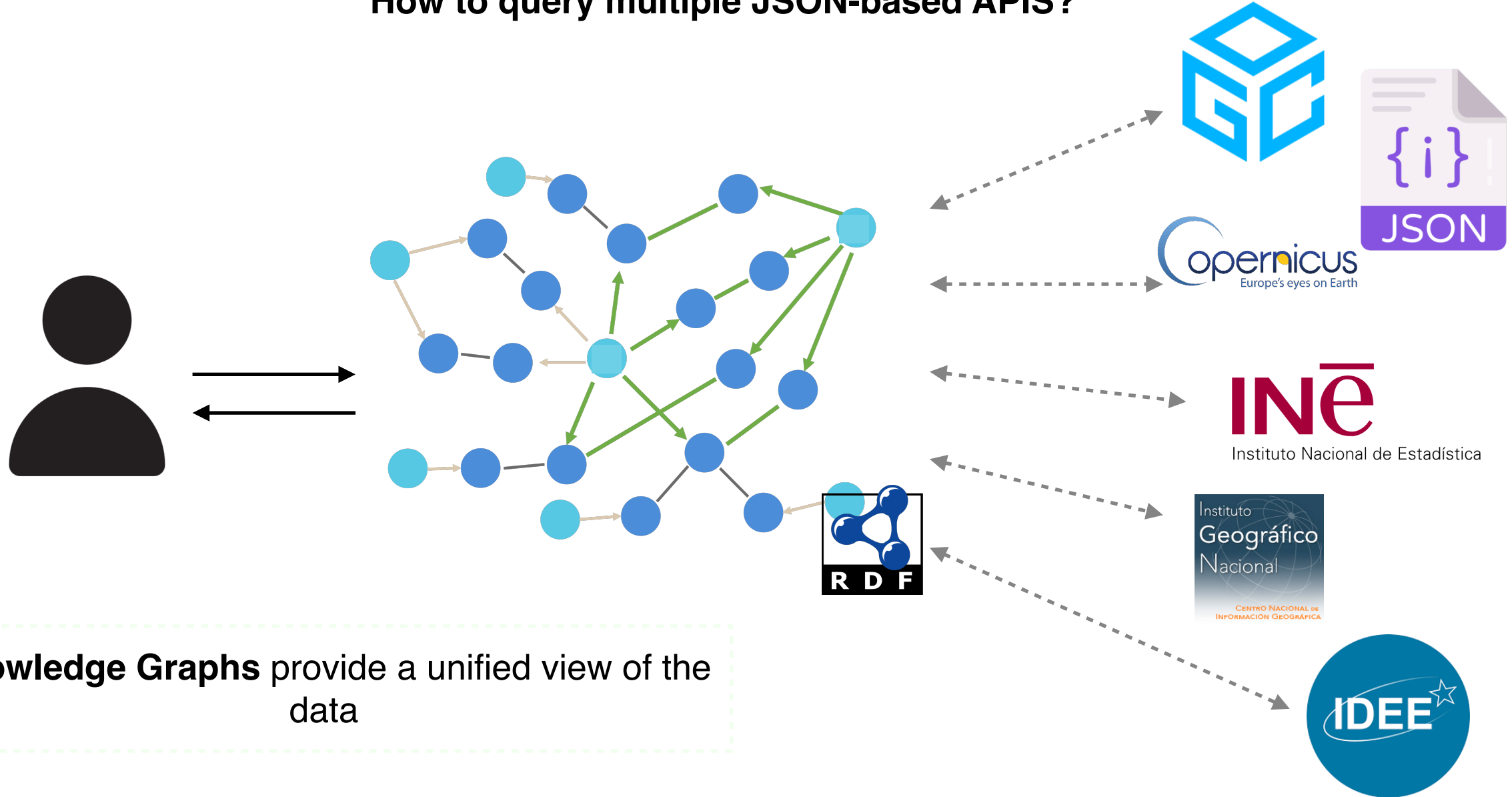
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Luis Manuel Vilches Blázquez
Oscar Corcho

How to query multiple JSON-based APIS?



How to query multiple JSON-based APIs?



Source 1

```
{
  "type": "Feature",
  "properties": {
    "fuente": "Instituto Geogr\u00e1fico Nacional",
    "nombre": "A Carrasqueira",
    "fecha_alta": "2019-04-09",
    "area": 118414.0,
    "id_demarc": "Galicia-Costa",
    "tipo": "Marisma",
  },
  "id": "ES014HYWLD50000001553534",
  "geometry": {
    "type": "Polygon",
    "coordinates": ...
  }
}
```

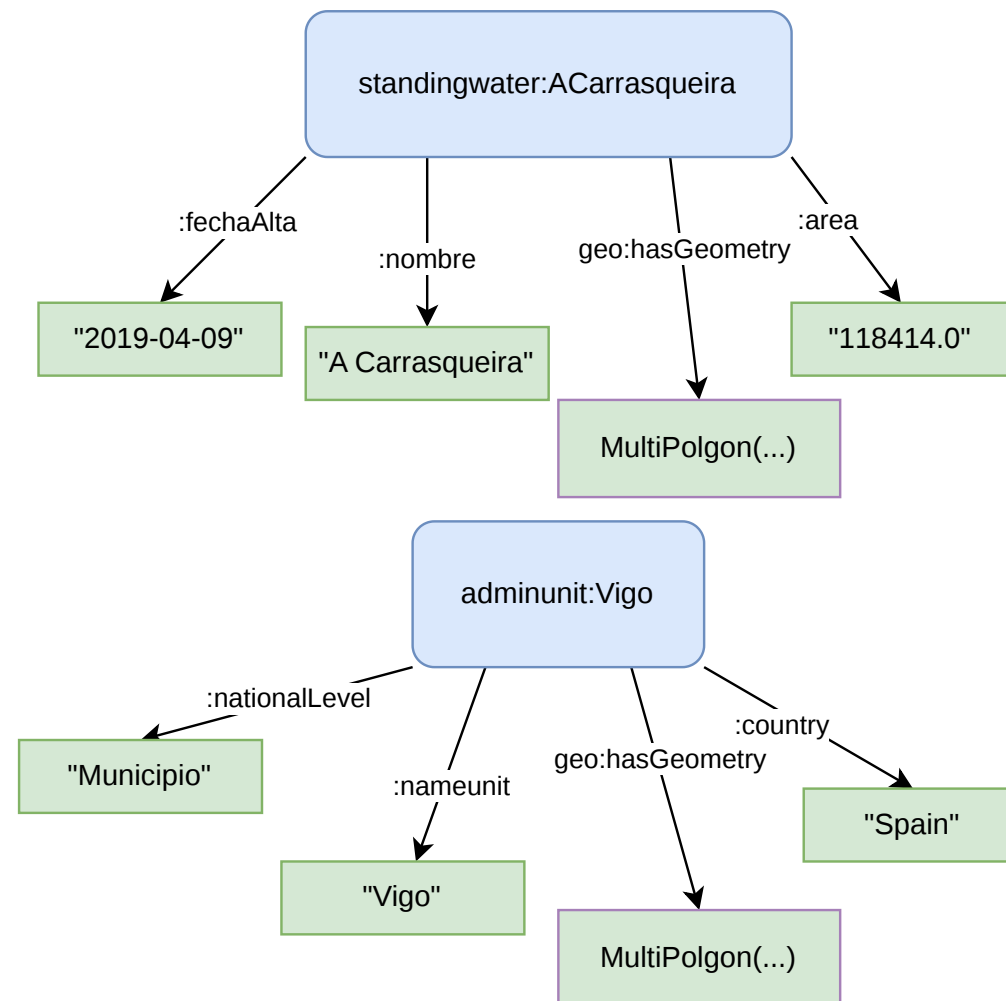


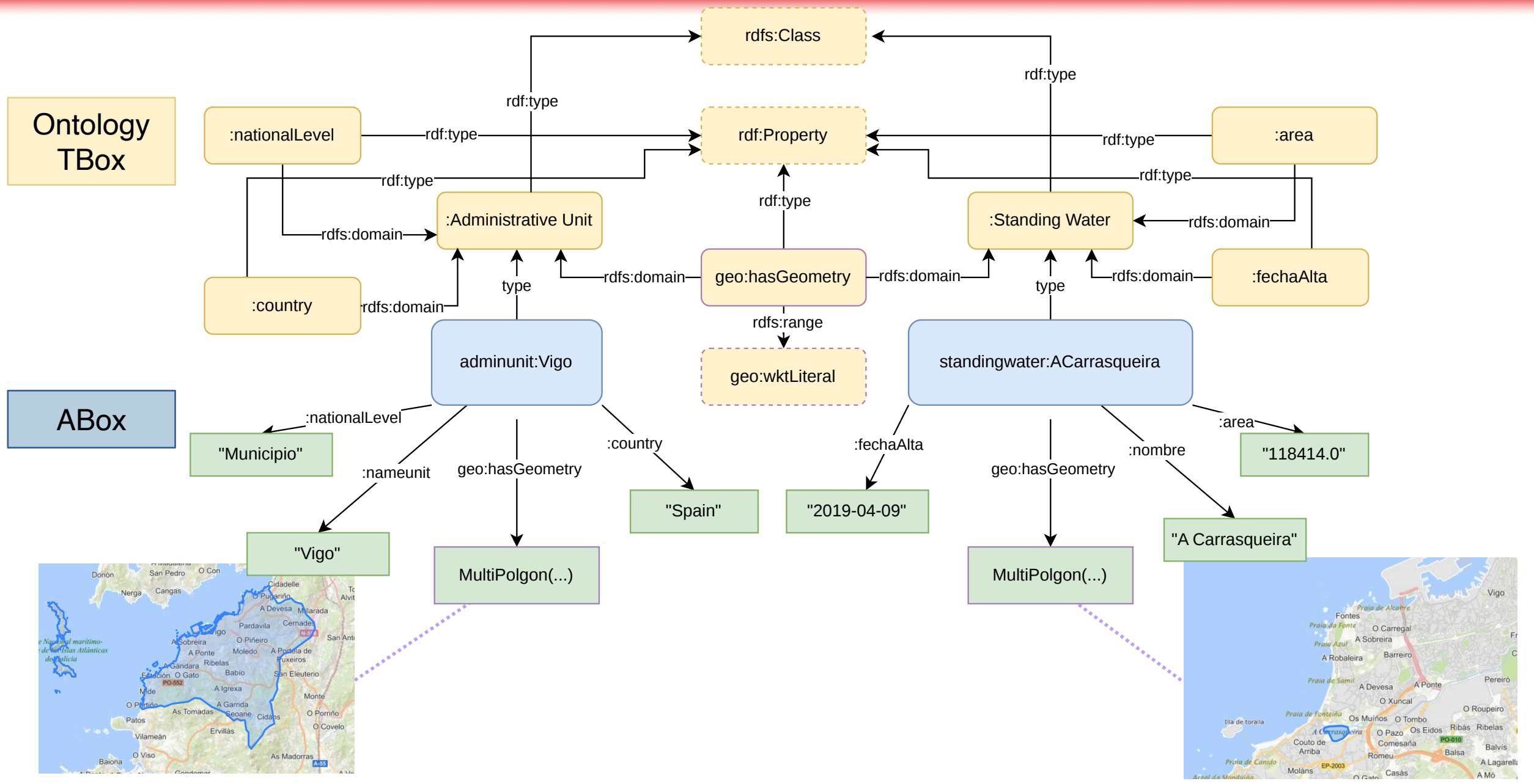
Source 2

```
{
  "type": "Feature",
  "properties": {
    "nameunit": "Vigo",
    "country": "ES",
    "nationalcode": "34123636057",
    "nationallevelname": "Municipio"
  },
  "id": "1118004",
  "geometry": {
    "type": "MultiPolygon",
    "coordinates": [ ...
  ]
}
```



=





How do we transform the data?

Source 1

```
{
  "type": "Feature",
  "properties": {
    "fuented": "Instituto Geogr\u00e1fico Nacional",
    "nombre": "A Carrasqueira",
    "fecha_alta": "2019-04-09",
    "area": 118414.0,
    "id_demarc": "Galicia-Costa",
    "tipo": "Marisma",
  },
  "id": "ES014HYWLD0000001553534",
  "geometry": {
    "type": "Polygon",
    "coordinates": ...
  }
}
```



```
:standingwaterTriplesMap a rml:TriplesMap ;
  rml:logicalSource :LogicalSource_administrativeunit ;

  rml:predicateObjectMap [ rml:objectMap [ void:filterx "bbox=@{1}" ;
    rml:datatype geo:geoJSONLiteral ;
    rml:reference "geometry" ] ;
    rml:predicate geo:hasGeometry ],

  [ rml:objectMap [ void:filterx "nombre=@{1}" ;
    rml:datatype xsd:string ;
    rml:reference "properties.nameunit" ] ;
    rml:predicate :nombre ] ;

  rml:subjectMap [ rml:class geolinkeddata:Aguas_Quietas ;
    rml:template "https://api-features.idee.es/.../standingwater/items/{id}" ] .
```



RML mappings are used to transform source data into RDF triples

Source 1

```
{
  "type": "Feature",
  "properties": {
    "fuented": "Instituto Geogr\u00e1fico Nacional",
    "nombre": "A Carrasqueira",
    "fecha_alta": "2019-04-09",
    "area": 118414.0,
    "id_demarc": "Galicia-Costa",
    "tipo": "Marisma",
    "id": "ES014HYWLD0000001553534",
    "geometry": {
      "type": "Polygon",
      "coordinates": ...
    }
  }
}
```

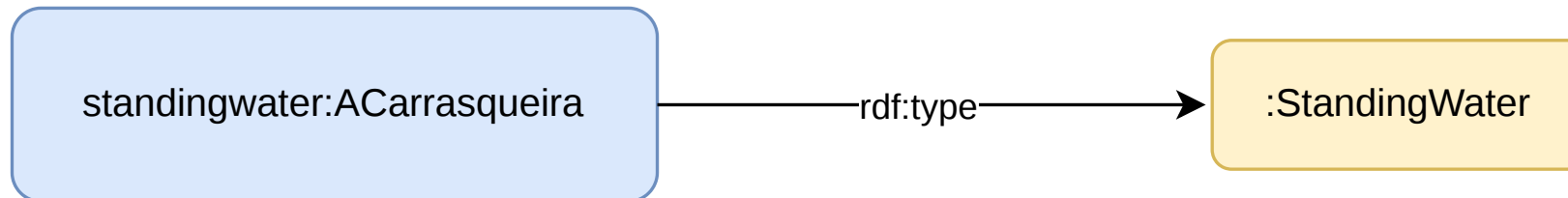


```
:standingwaterTriplesMap a rml:TriplesMap ;
  rml:logicalSource :LogicalSource_administrativeunit ;

  rml:predicateObjectMap [ rml:objectMap [ void:filterx "bbox=@{1}" ;
    rml:datatype geo:geoJSONLiteral ;
    rml:reference "geometry" ] ;
    rml:predicate geo:hasGeometry ],

  [ rml:objectMap [ void:filterx "nombre=@{1}" ;
    rml:datatype xsd:string ;
    rml:reference "properties.nameunit" ] ;
    rml:predicate :nombre ] ;

  rml:subjectMap [ rml:class geolinkeddata:Aguas_Quietas ;
    rml:template "https://api-features.idee.es/.../standingwater/items/{id}" ] .
```



```
<template("standingwater/{id}"), rdfs:type, URI(:StandingWater)>
```

Source 1

```
{
  "type": "Feature",
  "properties": {
    "fuented": "Instituto Geogr\u00e1fico Nacional",
    "nombre": "A Carrasqueira",
    "fecha_alta": "2019-04-09",
    "area": 118414.0,
    "id_demarc": "Galicia-Costa",
    "tipo": "Marisma",
  },
  "id": "ES014HYWLD0000001553534",
  "geometry": {
    "type": "Polygon",
    "coordinates": ...
  }
}
```

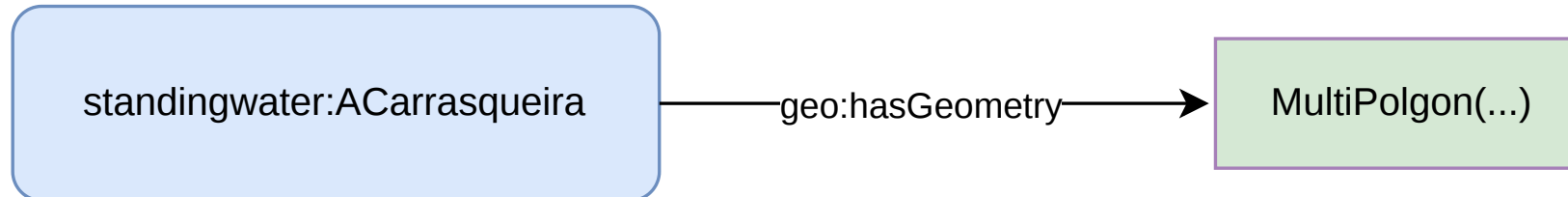


```
:standingwaterTriplesMap a rml:TriplesMap ;
  rml:logicalSource :LogicalSource_administrativeunit ;

  rml:predicateObjectMap [ rml:objectMap [ void:filterx "bbox=@{1}" ;
    rml:datatype geo:geoJSONLiteral ;
    rml:reference "geometry" ] ;
    rml:predicate geo:hasGeometry ],

  [ rml:objectMap [ void:filterx "nombre=@{1}" ;
    rml:datatype xsd:string ;
    rml:reference "properties.nameunit" ] ;
    rml:predicate :nombre ] ;

  rml:subjectMap [ rml:class geolinkeddata:Aguas_Quietas ;
    rml:template "https://api-features.idee.es/.../standingwater/items/{id}" ] .
```



```
<template(" :standingwater/{id}" ), geo:hasGeometry, Literal(MultiPolygon(...))>
```

Source 1

```
{
  "type": "Feature",
  "properties": {
    "fuente": "Instituto Geográfico Nacional",
    "nombre": "A Carrasqueira",
    "fecha_alta": "2019-04-09",
    "area": 118414.0,
    "id_demarc": "Galicia-Costa",
    "tipo": "Marisma",
  },
  "id": "ES014HYWLD0000001553534",
  "geometry": {
    "type": "Polygon",
    "coordinates": ...
  }
}
```



```
:standingwaterTriplesMap a rml:TriplesMap ;
  rml:logicalSource :LogicalSource_administrativeunit ;

  rml:predicateObjectMap [ rml:objectMap [ void:filterx "bbox=@{1}" ;
    rml:datatype geo:geoJSONLiteral ;
    rml:reference "geometry" ] ;
    rml:predicate geo:hasGeometry ],

  [ rml:objectMap [ void:filterx "nombre=@{1}" ;
    rml:datatype xsd:string ;
    rml:reference "properties.nameunit" ] ;
    rml:predicate :nombre ] ;

  rml:subjectMap [ rml:class geolinkeddata:Aguas_Quietas ;
    rml:template "https://api-features.idee.es/.../standingwater/items/{id}" ] .
```



standingwater:ACarrasqueira

:nombre

"A Carrasqueira"

```
<template(" :standingwater/{id}"), :nombre, reference("properties.nameunit")>
```

How do we query the data?



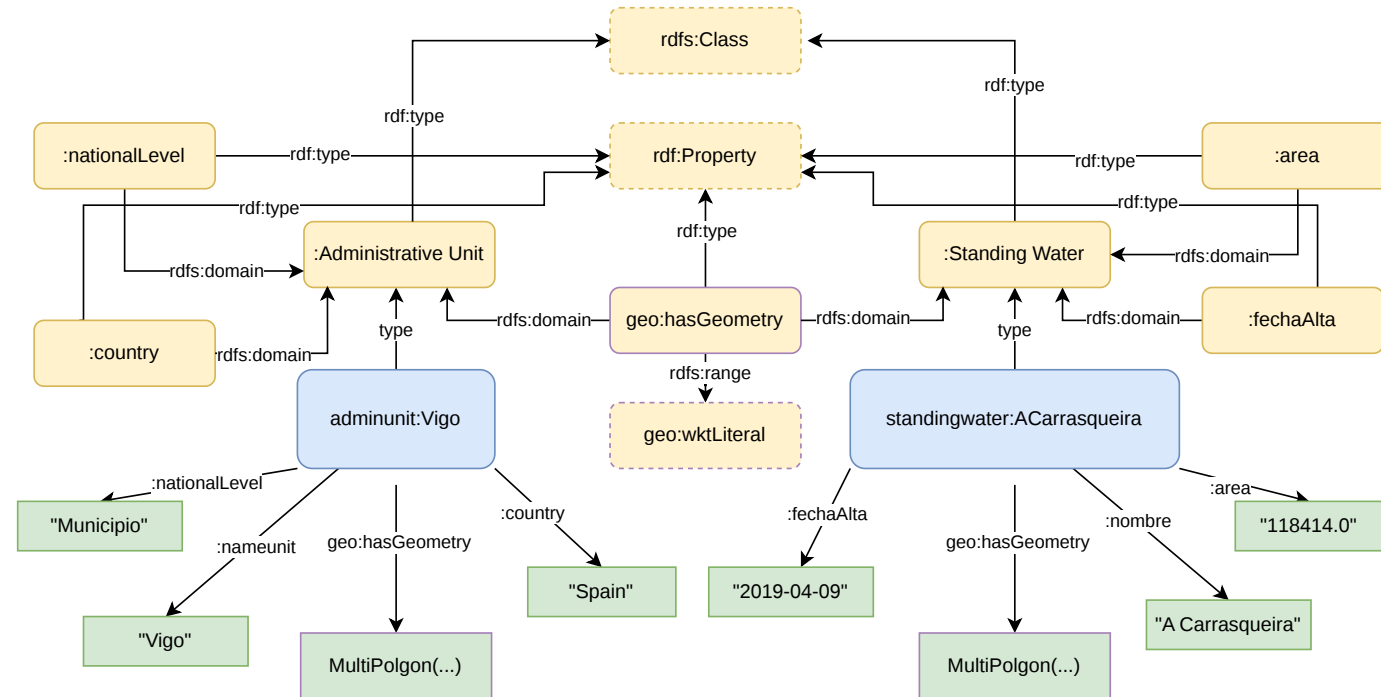
```

SELECT ?x ?a ?gx WHERE{
  ?v a :administrativeunit ;
    :nameunit "Vigo" ;
    geo:hasGeometry ?gv .

  ?x a :standingwater ;
    :area ?a ;
    geo:hasGeometry ?gx .

  FILTER ( geof:sfContains(?gv, ?gx) )
}

```



How do we query the data?



```

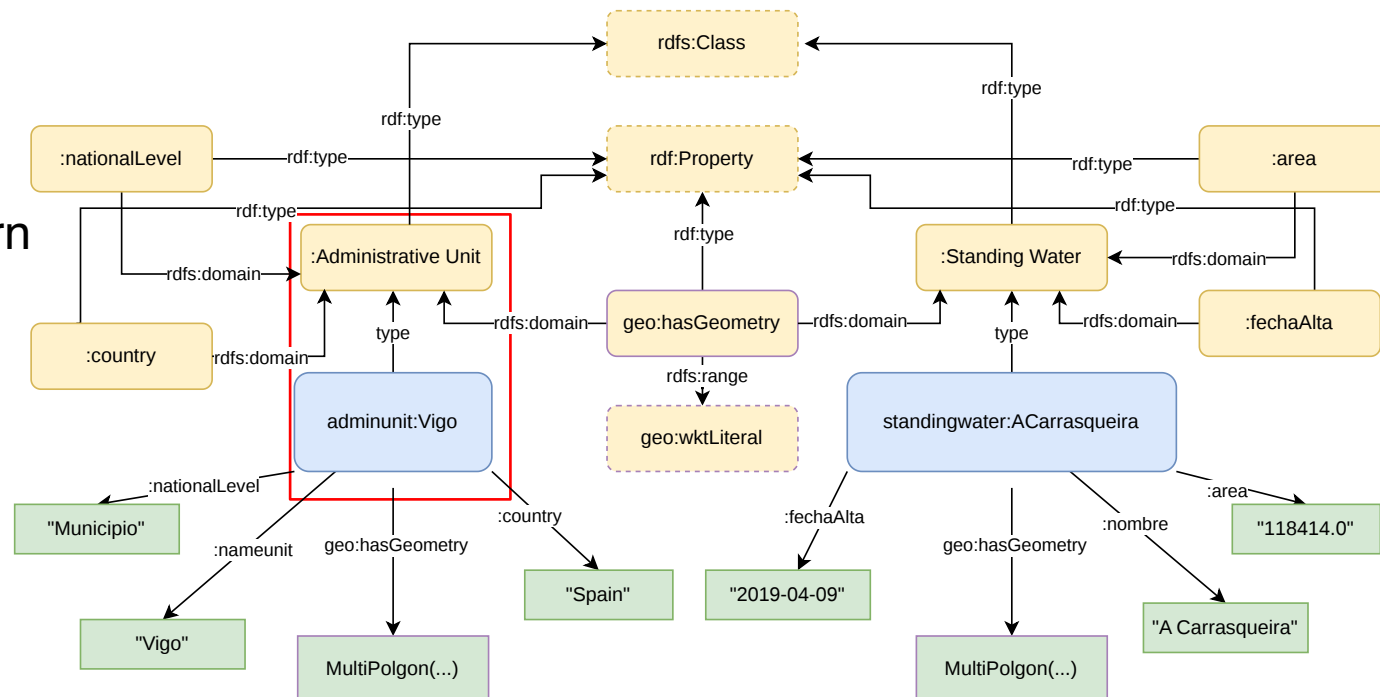
SELECT ?x ?a ?gx WHERE{
  ?v a :administrativeunit ;
    :nameunit "Vigo" ;
    geo:hasGeometry ?gv .

  ?x a :standingwater ;
    :area ?a ;
    geo:hasGeometry ?gx .

  FILTER ( geof:sfContains(?gv, ?gx) )
}

```

Graph Pattern



How do we query the data?



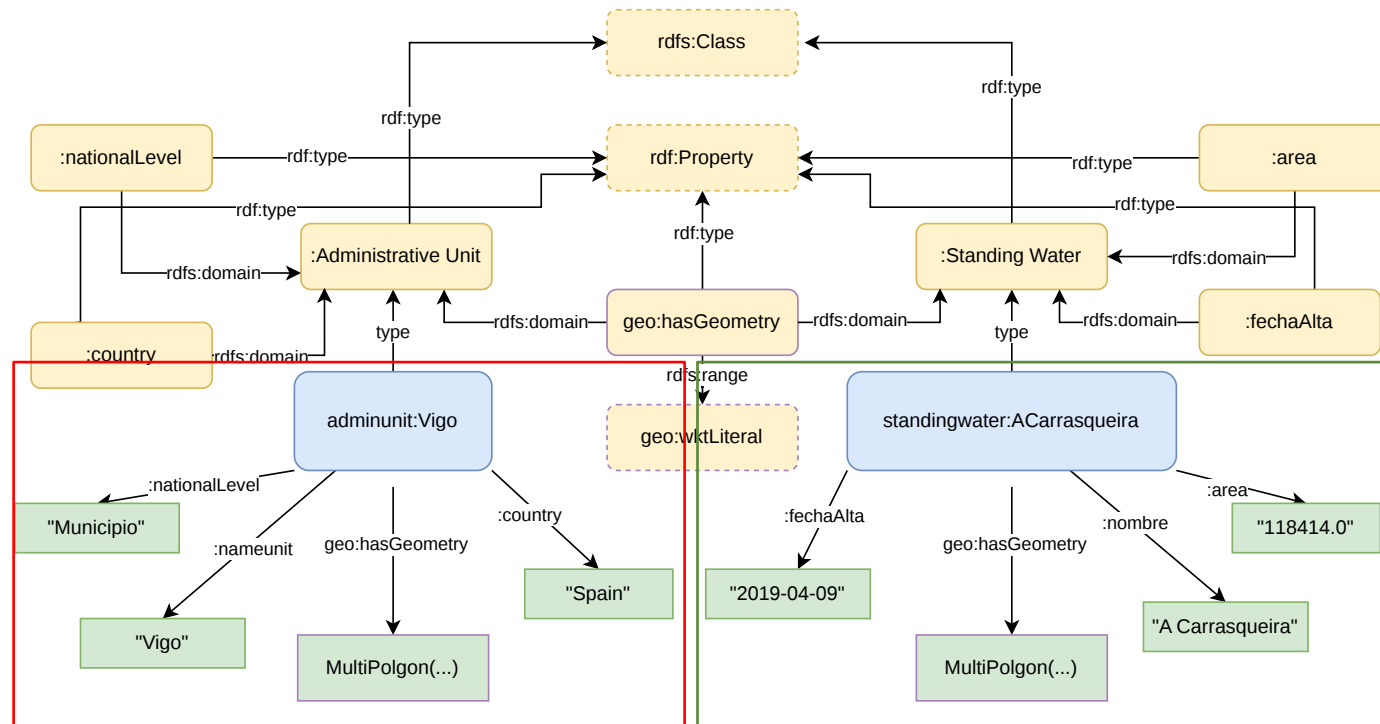
```
SELECT ?x ?a ?gx WHERE{
```

```
  ?v a :administrativeunit ;  
    :nameunit "Vigo" ;  
    geo:hasGeometry ?gv .
```

```
  ?x a :standingwater ;  
    :area ?a ;  
    geo:hasGeometry ?gx .
```

```
FILTER ( geof:sfContains(?gv, ?gx) )  
}
```

Star-Shaped
Basic Graph
Patterns



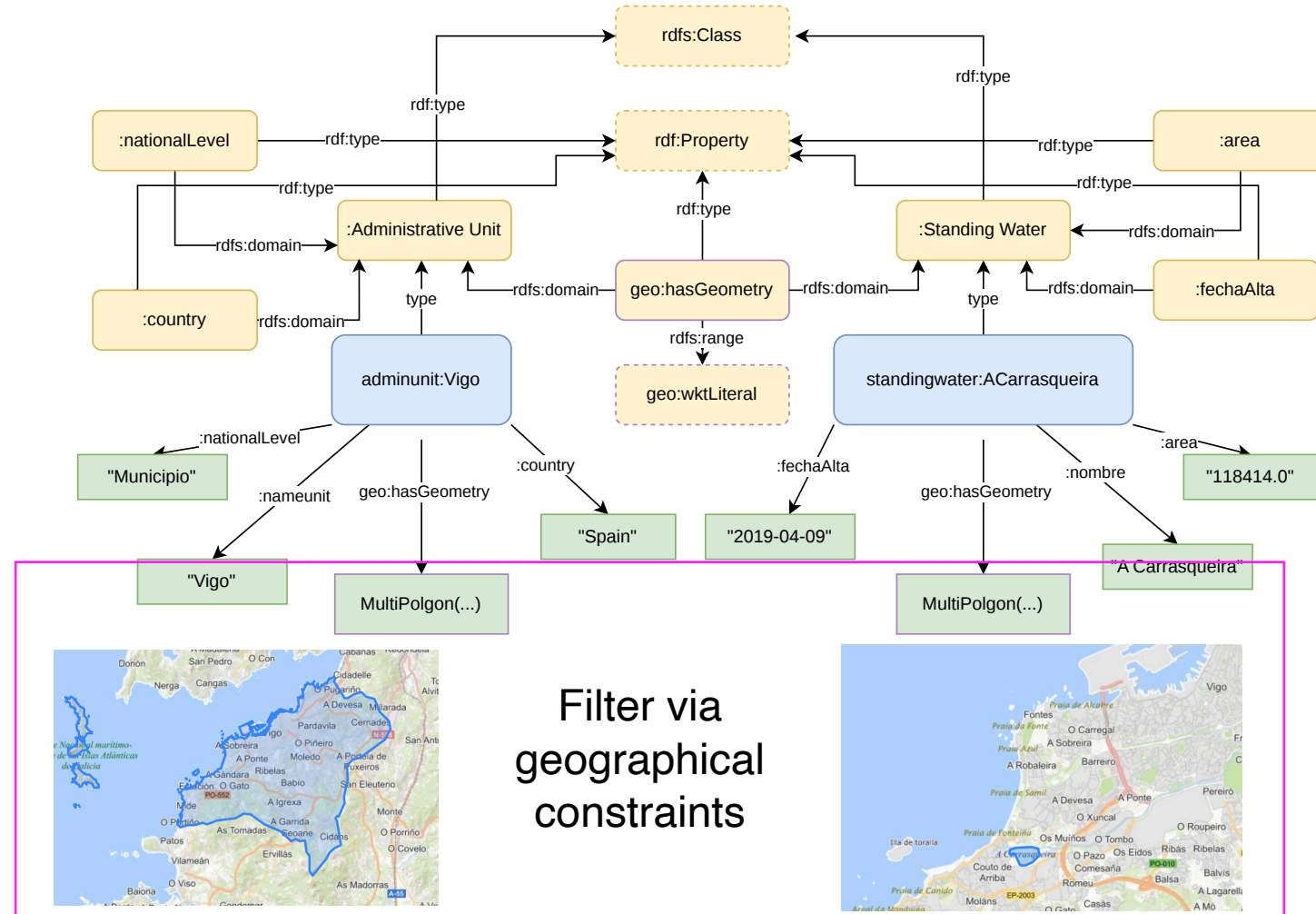
How do we query the data?



```
SELECT ?x ?a ?gx WHERE{
  ?v a :administrativeunit ;
    :nameunit "Vigo" ;
    geo:hasGeometry ?gv .
```

```
  ?x a :standingwater ;
    :area ?a ;
    geo:hasGeometry ?gx .
```

```
FILTER ( geof:sfContains(?gv, ?gx) )
}
```



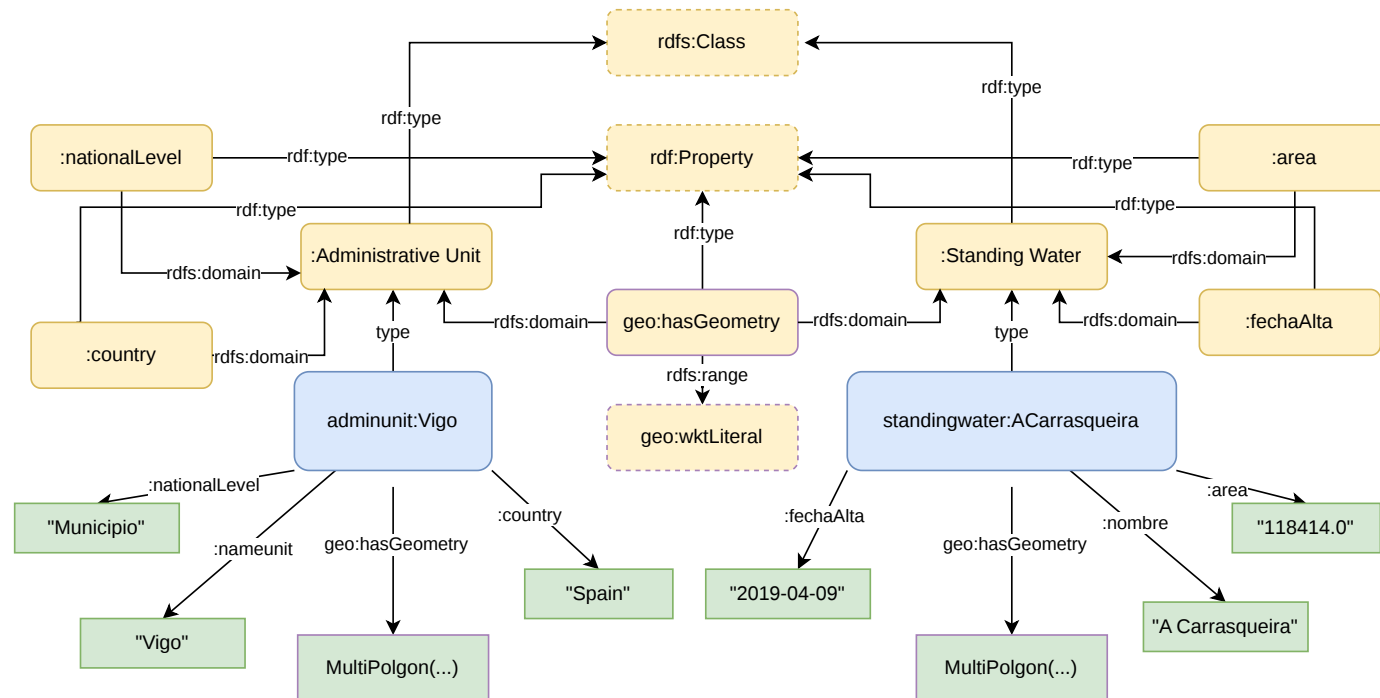
How do we query the data?



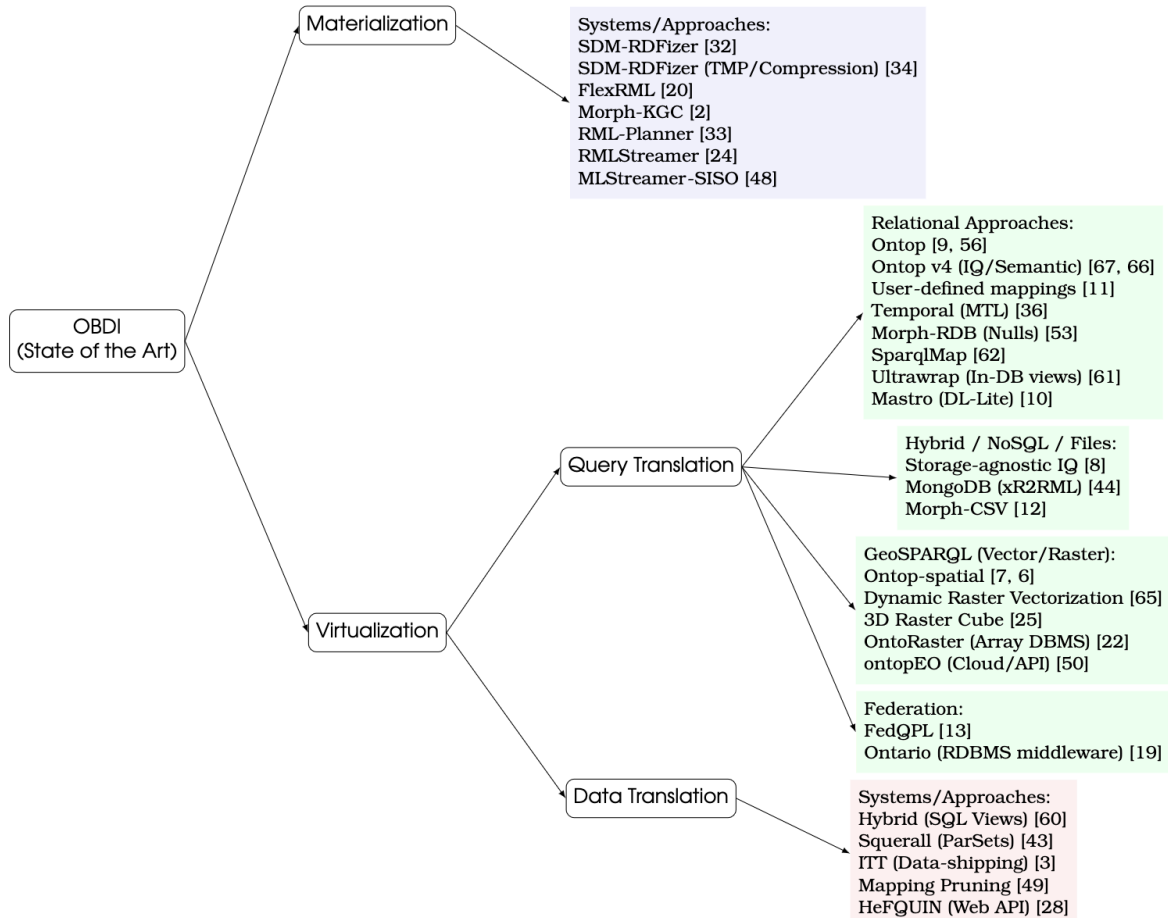
```
SELECT ?x ?a ?gx WHERE{
  ?v a :administrativeunit ;
    :nameunit "Vigo" ;
    geo:hasGeometry ?gv .

  ?x a :standingwater ;
    :area ?a ;
    geo:hasGeometry ?gx .

  FILTER ( geof:sfContains(?gv, ?gx) )
}
```



?x → standingwater:ACarrasqueira
 ?a → "118414.0"
 ?gx → MultiPolygon(...)



Materialization

Materialize the complete sources onto RDF and the query

Duplicated data

High materialization cost in every source update

Query Translation

Translate SPARQL query onto source query language

Expressive intermediate JSON query language

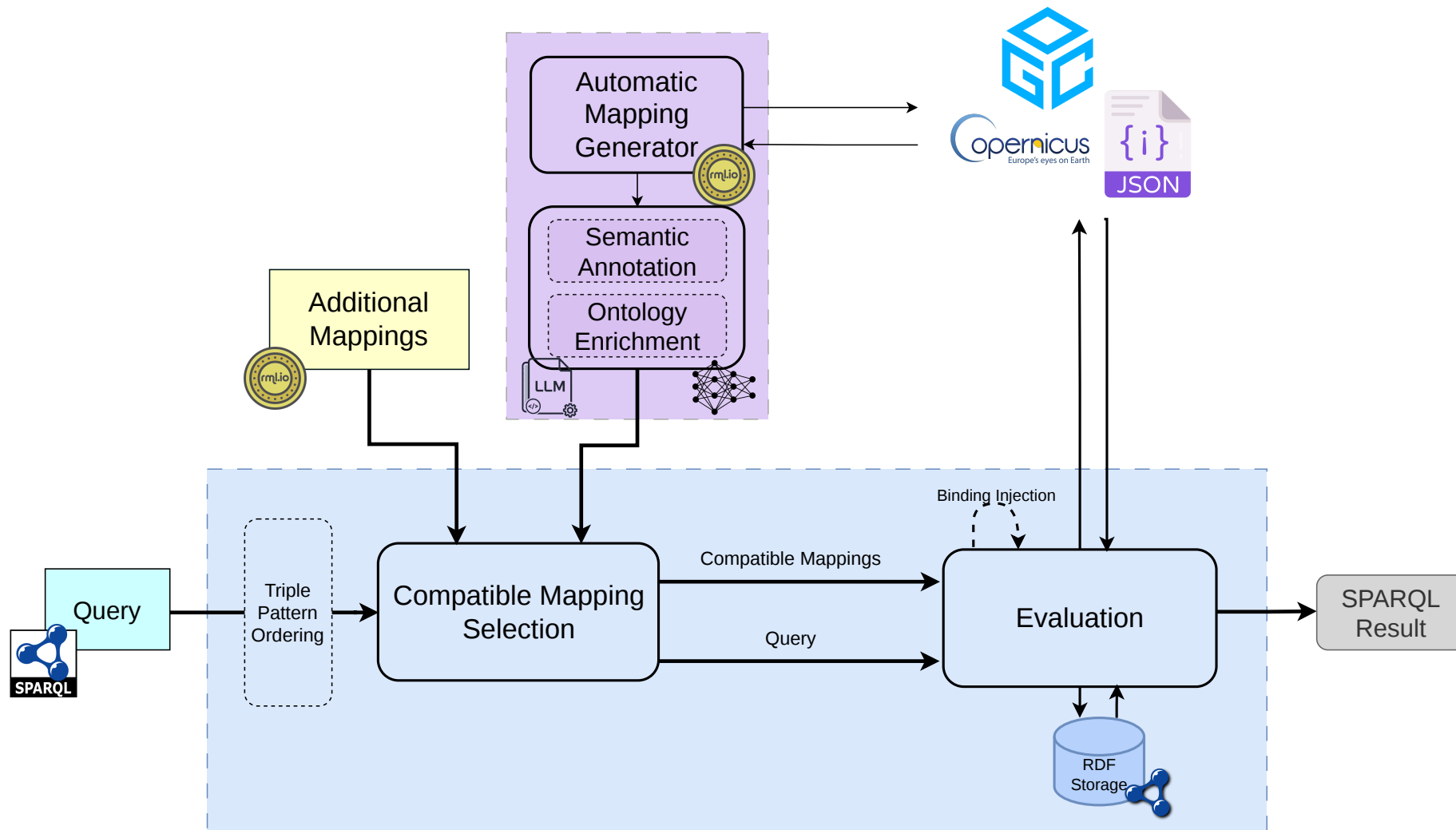
Hard to interoperate heterogenous sources

Data Translation

Materialize the minimum RDF triples in order to satisfy a given SPARQL query

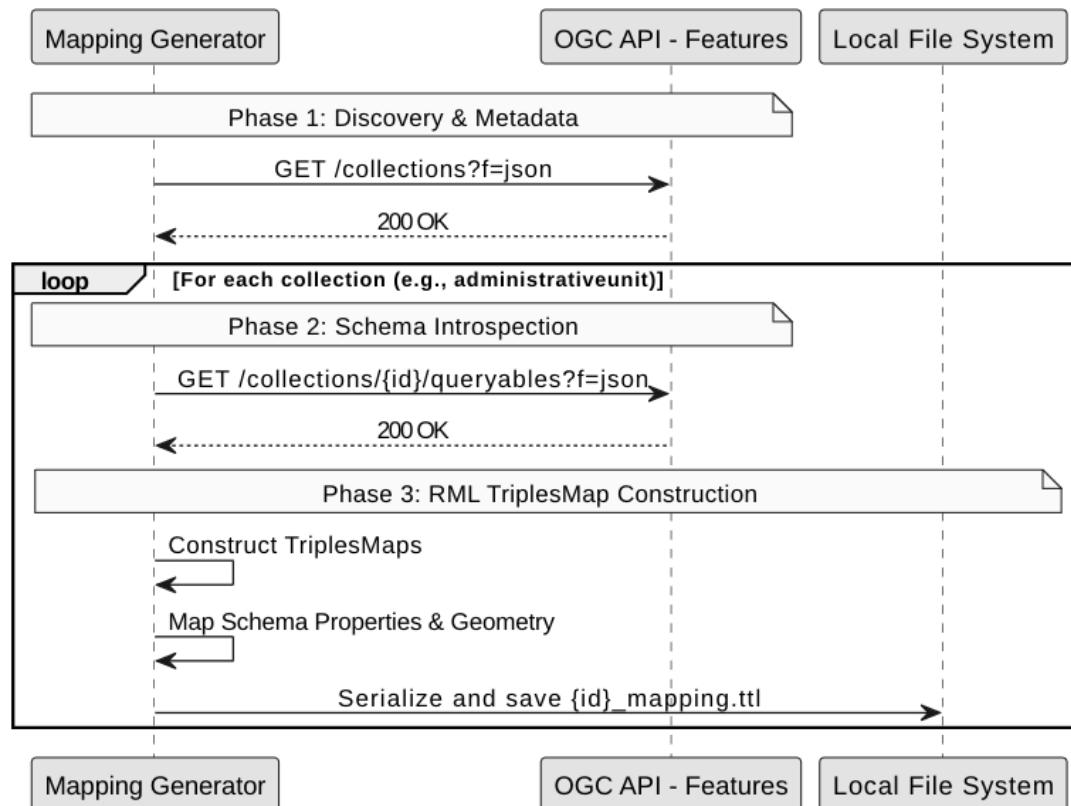
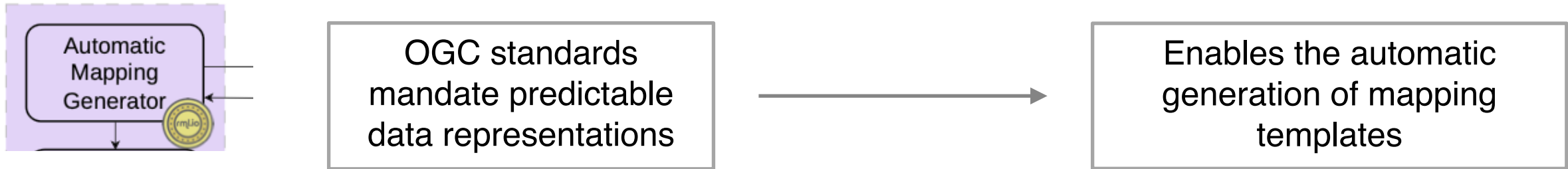
Schema agnostic

Materialize only what you need



- Automatically generated mapping templates
- LLM-driven semantic annotation and ontology enrichment

- Materialization engine that materializes minimum mapping rules given a query.
- Optimizes performance by injecting bindings and pushing down filters directly into the HTTP



Retrieve collections of data the API offers

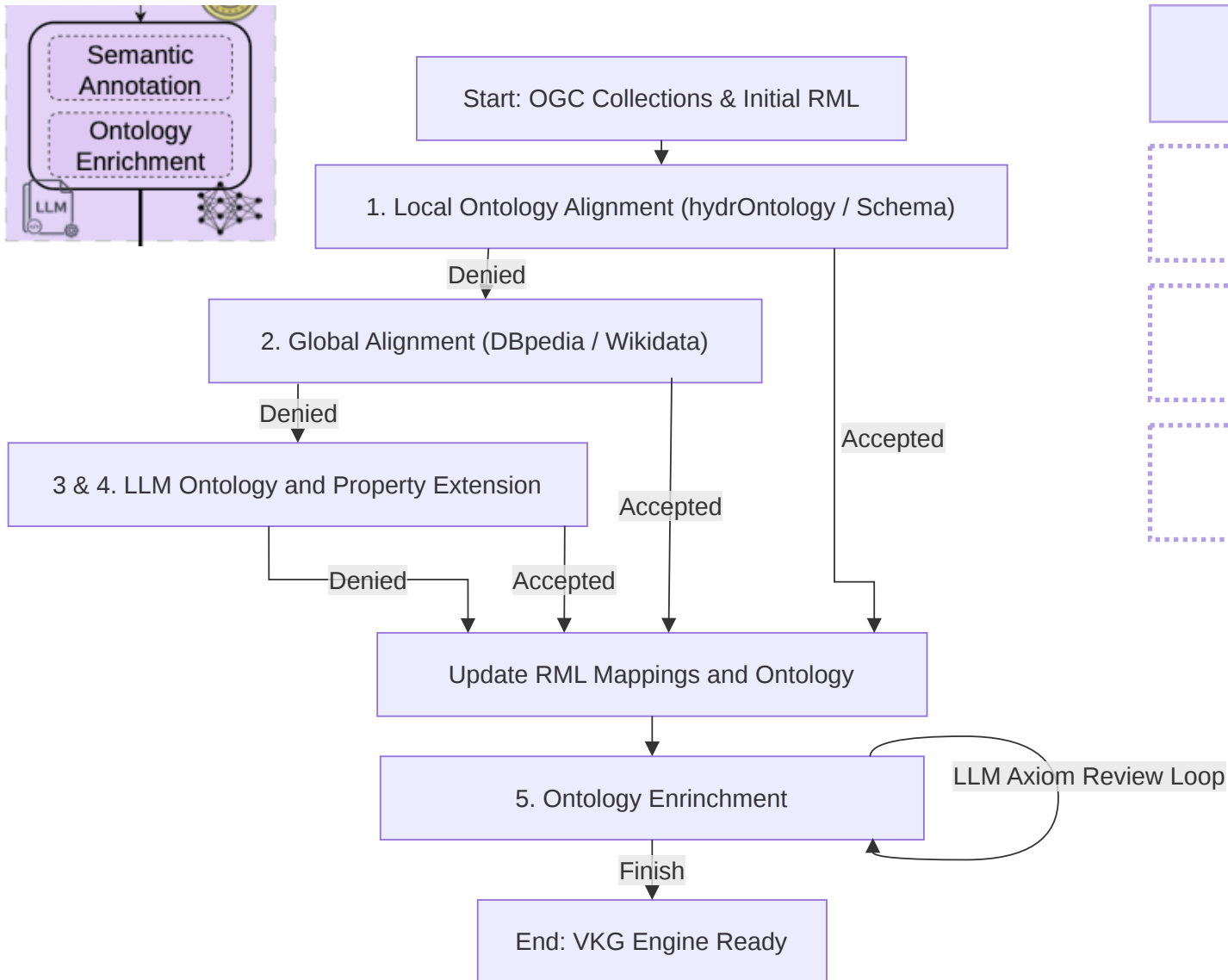
e.g. falls, locks, wetland, standingwaters...

Retrieve internal schema of every collection

For standingwaters: geometry, nombre, area, elevation...

Generate Mapping Templates

RML TriplesMaps



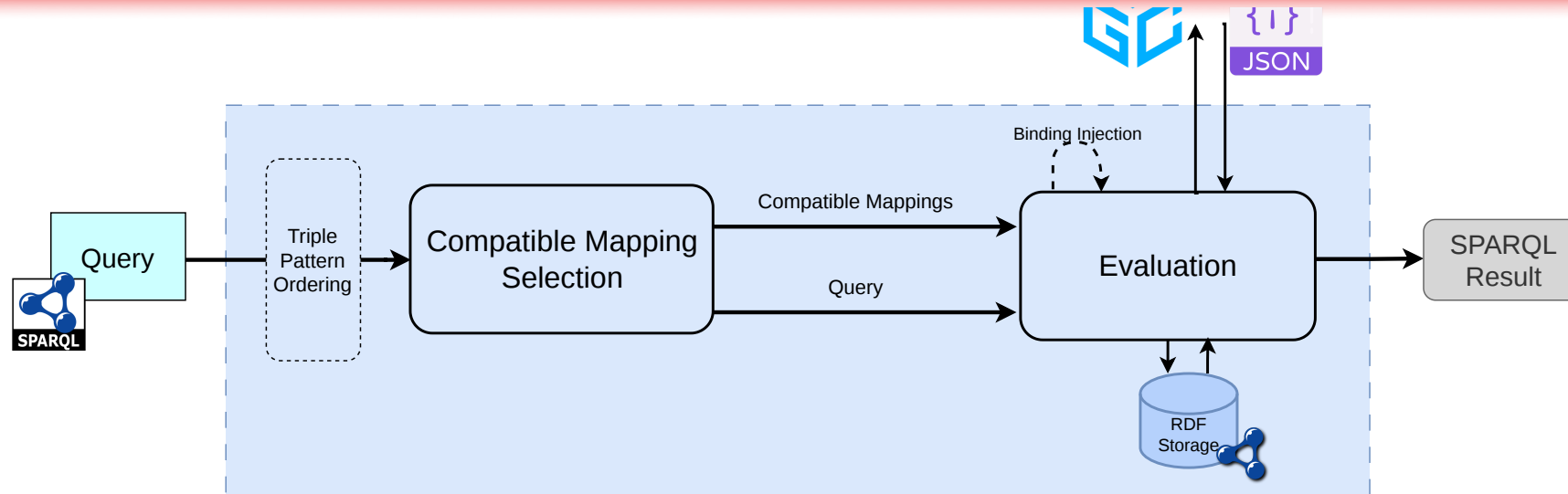
Human-in-the-loop approach

Cosine Similarity between classes and properties embeddings

LLM proposes DBpedia and Wikidata URIs
Checks existence with SPARQL query

LLM proposes a new class extension as a subclass

LLM iteratively proposes ontology axioms.
(owl:someValuesFrom,
owl:allValuesFrom, owl:cardinality...)



```

SELECT ?w ?gw WHERE {
  ?w a :agua:Zif_riesgo_pob_q500 ;
    geo:hasGeometry ?gw .
  ?s a :administrativeunit ;
    geolinkeddata:nombre "Lugo" ;
    :nationallevelname "Provincia" ;
    geo:hasGeometry ?gs .
  ?t a geolinkeddata:Humedal ;
    geo:hasGeometry ?gt .
  FILTER (geof:sfWithin(?gw, ?gs))
  FILTER (geof:sfWithin(?gt, ?gs))
  FILTER (geof:sfDistance(?gw, ?gt) < 2000)
}

```

Get the **flooding risk areas** in **Lugo** that are less than 2000m away from a **wetland**

Mapping Selection

<code>?s a :administrativeunit .</code>	<code>?s geo:hasGeometry ?gs .</code>	<code>?s :nationallevelname "Provincia"</code>
---	---------------------------------------	--

```
<template(":adminunit/{id}"), rdfs:type, URI(:administrativeunit)>  
<template(":adminunit/{id}"), :nationallevelname, reference("$.nationallevelname")>  
<template(":standingwater/{id}"), geo:hasGeometry, reference("$.geometry")>  
<template(":adminunit/{id}"), geo:hasGeometry, reference("$.geometry")>
```

Mapping Selection

`?s a :administrativeunit .` `?s geo:hasGeometry ?gs .` `?s :nationallevelname "Provincia"`



```

<template(":adminunit/{id}"), rdfs:type, URI(:administrativeunit)>
<template(":adminunit/{id}"), :nationallevelname, reference("$.nationallevelname")>
<template(":standingwater/{id}"), geo:hasGeometry, reference("$.geometry")>
<template(":adminunit/{id}"), geo:hasGeometry, reference("$.geometry")>
  
```

`?s → template(":adminunit/{id}")`

Mapping Selection

<code>?s a :administrativeunit .</code>	<code>?s geo:hasGeometry ?gs .</code>	<code>?s :nationallevelname "Provincia"</code>
---	---------------------------------------	--



```
<template(":adminunit/{id}"), rdfs:type, URI(:administrativeunit)>
```

```
<template(":adminunit/{id}"), :nationallevelname, reference("$.nationallevelname")>
```

```
<template(":standingwater/{id}"), geo:hasGeometry, reference("$.geometry")>
```

```
<template(":adminunit/{id}"), geo:hasGeometry, reference("$.geometry")>
```

```
?s -> template(":adminunit/{id}")
```

Mapping Selection

<code>?s a :administrativeunit .</code>	<code>?s geo:hasGeometry ?gs .</code>	<code>?s :nationallevelname "Provincia"</code>
---	---------------------------------------	--



```
<template(":adminunit/{id}"), rdfs:type, URI(:administrativeunit)>
```

```
<template(":adminunit/{id}"), :nationallevelname, reference("$.nationallevelname")>
```

```
<template(":standingwater/{id}"), geo:hasGeometry, reference("$.geometry")>
```

```
<template(":adminunit/{id}"), geo:hasGeometry, reference("$.geometry")>
```

`?s → template(":adminunit/{id}")`

Mapping Selection

`?s a :administrativeunit .` `?s geo:hasGeometry ?gs .` `?s :nationallevelname "Provincia"`

`<template(":adminunit/{id}"), rdfs:type, URI(":administrativeunit")>`

`<template(":adminunit/{id}"), :nationallevelname, reference("$.nationallevelname")>`

`<template(":standingwater/{id}"), geo:hasGeometry, reference("$.geometry")>`

`<template(":adminunit/{id}"), geo:hasGeometry, reference("$.geometry")>`

`?s → template(":adminunit/{id}")`

Mapping Selection

`?s a :administrativeunit .` `?s geo:hasGeometry ?gs .` `?s :nationallevelname "Provincia"`

`<template(":adminunit/{id}"), rdfs:type, URI(:administrativeunit)>`

`<template(":adminunit/{id}"), :nationallevelname, reference("$.nationallevelname")>`

`<template(":standingwater/{id}"), geo:hasGeometry, reference("$.geometry")>`

`<template(":adminunit/{id}"), geo:hasGeometry, reference("$.geometry")>`

`?s → template(":adminunit/{id}")`

Mapping Selection

<code>?s a :administrativeunit .</code>	<code>?s geo:hasGeometry ?gs .</code>	<code>?s :nationallevelname "Provincia"</code>
---	---------------------------------------	--

```
<template(":adminunit/{id}"), rdfs:type, URI(:administrativeunit)>
```

```
<template(":adminunit/{id}"), :nationallevelname, reference("$.nationallevelname")>
```

```
<template(":standingwater/{id}"), geo:hasGeometry, reference("$.geometry")>
```

```
<template(":adminunit/{id}"), geo:hasGeometry, reference("$.geometry")>
```

`?s → template(":adminunit/{id}")`

Mapping Selection

What if...?

<code>?s a :administrativeunit .</code>	<code>?s geo:hasGeometry ?gs .</code>	<code>?s :nationallevelname "Provincia"</code>
<code>?s :notexist ?a</code>		

```
<template(":adminunit/{id}"), rdfs:type, URI(:administrativeunit)>
```

```
<template(":adminunit/{id}"), :nationallevelname, reference("$.nationallevelname")>
```

```
<template(":standingwater/{id}"), geo:hasGeometry, reference("$.geometry")>
```

```
<template(":adminunit/{id}"), geo:hasGeometry, reference("$.geometry")>
```



`?s → template(":adminunit/{id}")`

**No mapping
group compatible**

Star-Shaped Mapping Groups

```

SELECT ?w ?gw WHERE {
  ?w a :agua:Zif_riesgo_pob_q500 ;
     geo:hasGeometry ?gw .
  ?s a :administrativeunit ;
     geolinkeddata:nombre "Lugo" ;
     :nationallevelname "Provincia" ;
     geo:hasGeometry ?gs .
  ?t a geolinkeddata:Humedal ;
     geo:hasGeometry ?gt .
  FILTER (geof:sfWithin(?gw, ?gs))
  FILTER (geof:sfWithin(?gt, ?gs))
  FILTER (geof:sfDistance(?gw, ?gt) < 2000)
}

```



Each Star-Shaped Mapping Group shares subject term map,
source and bounded variable

Heuristic Triple Pattern Ordering

Execute the most constrained Star-Shaped Basic Graph Pattern first so later queries can be filtered down

Constant URI or Literal in the object
+10

Geographical constrained variable
+20

If that variable is constrained to a literal geometry
+10 (+30 overall)

```
SELECT ?w ?gw WHERE {
  ?w a :agua:Zif_riesgo_pob_q500 ;
      geo:hasGeometry ?gw .
  ?s a :administrativeunit ;
      geolinkeddata:nombre "Lugo" ;
      :nationallevelname "Provincia" ;
      geo:hasGeometry ?gs .
  ?t a geolinkeddata:Humedal ;
      geo:hasGeometry ?gt .
  FILTER (geof:sfWithin(?gw, ?gs))
  FILTER (geof:sfWithin(?gt, ?gs))
  FILTER (geof:sfDistance(?gw, ?gt) < 2000)
}
```

+20

+40 This should be queried first

+20

Query Evaluation

```

SELECT ?w ?gw WHERE {
  ?w a :agua:Zif_riesgo_pob_q500 ;
     geo:hasGeometry ?gw .
  ?s a :administrativeunit ;
     geolinkeddata:nombre "Lugo" ;
     :nationallevelname "Provincia" ;
     geo:hasGeometry ?gs .
  ?t a geolinkeddata:Humedal ;
     geo:hasGeometry ?gt .
  FILTER (geof:sfWithin(?gw, ?gs))
  FILTER (geof:sfWithin(?gt, ?gs))
  FILTER (geof:sfDistance(?gw, ?gt) < 2000)
}

```

<https://api-features.ign.es/collections/administrativeunit/items>
?nameunit="Lugo"
?nationallevelname="Provincia"

First triple pattern of SBGP
acts as a materialization
trigger



Query Evaluation

```

SELECT ?w ?gw WHERE {
  ?w a :agua:Zif_riesgo_pob_q500 ;
  geo:hasGeometry ?gw .
  ?s a :administrativeunit ;
  geolinkeddata:nombre "Lugo" ;
  :nationallevelname "Provincia" ;
  geo:hasGeometry ?gs .
  ?t a geolinkeddata:Humedal ;
  geo:hasGeometry ?gt .
  FILTER (geof:sfWithin(?gw, ?gs))
  FILTER (geof:sfWithin(?gt, ?gs))
  FILTER (geof:sfDistance(?gw, ?gt) < 2000)
}

```

https://api-features.ign.es/collections/administrativeunit/items
 ?nameunit="Lugo"
 ?nationallevelname="Provincia"



```

administrativeunit:Lugo a :administrativeunit .
administrativeunit:Lugo geolinkeddata:nombre "Lugo" .
administrativeunit:Lugo :nationallevelname "Provincia" .
administrativeunit:Lugo hasGeometry "POLYGON(...)" .

```

?s → administrativeunit:Lugo

Query Evaluation

```

SELECT ?w ?gw WHERE {
  ?w a :agua:Zif_riesgo_pob_q500 ;
    geo:hasGeometry ?gw .
  ?s a :administrativeunit ;
    geolinkeddata:nombre "Lugo" ;
    :nationallevelname "Provincia" ;
    geo:hasGeometry ?gs .
  ?t a geolinkeddata:Humedal ;
    geo:hasGeometry ?gt .
  FILTER (geof:sfWithin(?gw, ?gs))
  FILTER (geof:sfWithin(?gt, ?gs))
  FILTER (geof:sfDistance(?gw, ?gt) < 2000)
}

```

https://api-features.ign.es/collections/administrativeunit/items
 ?nameunit="Lugo"
 ?nationallevelname="Provincia"



```

administrativeunit:Lugo a :administrativeunit .
administrativeunit:Lugo geolinkeddata:nombre "Lugo" .
administrativeunit:Lugo :nationallevelname "Provincia" .
administrativeunit:Lugo hasGeometry "POLYGON(...)" .

```

?s → administrativeunit:Lugo

Query Evaluation

```

SELECT ?w ?gw WHERE {
  ?w a :agua:Zif_riesgo_pob_q500 ;
    geo:hasGeometry ?gw .
  ?s a :administrativeunit ;
    geolinkeddata:nombre "Lugo" ;
    :nationallevelname "Provincia" ;
    geo:hasGeometry ?gs .
  ?t a geolinkeddata:Humedal ;
    geo:hasGeometry ?gt .
  FILTER (geof:sfWithin(?gw, ?gs))
  FILTER (geof:sfWithin(?gt, ?gs))
  FILTER (geof:sfDistance(?gw, ?gt) < 2000)
}

```

https://api-features.ign.es/collections/administrativeunit/items
 ?nameunit="Lugo"
 ?nationallevelname="Provincia"

administrativeunit:Lugo a :administrativeunit .
 administrativeunit:Lugo geolinkeddata:nombre "Lugo" .
 administrativeunit:Lugo :nationallevelname "Provincia" .
 administrativeunit:Lugo hasGeometry "POLYGON(...)" .

?s → administrativeunit:Lugo

Query Evaluation

```

SELECT ?w ?gw WHERE {
  ?w a :agua:Zif_riesgo_pob_q500 ;
    geo:hasGeometry ?gw .
  ?s a :administrativeunit ;
    geolinkeddata:nombre "Lugo" ;
    :nationallevelname "Provincia" ;
    geo:hasGeometry ?gs .
  ?t a geolinkeddata:Humedal ;
    geo:hasGeometry ?gt .
  FILTER (geof:sfWithin(?gw, ?gs))
  FILTER (geof:sfWithin(?gt, ?gs))
  FILTER (geof:sfDistance(?gw, ?gt) < 2000)
}

```

https://api-features.ign.es/collections/administrativeunit/items
 ?nameunit="Lugo"
 ?nationallevelname="Provincia"



```

administrativeunit:Lugo a :administrativeunit .
administrativeunit:Lugo geolinkeddata:nombre "Lugo" .
administrativeunit:Lugo :nationallevelname "Provincia" .
administrativeunit:Lugo hasGeometry "POLYGON(...)" .

```

?s → administrativeunit:Lugo
 ?gs → "POLYGON(...)"

Query Evaluation

```

SELECT ?w ?gw WHERE {
  ?w a :agua:Zif_riesgo_pob_q500 ;
     geo:hasGeometry ?gw .
  ?s a :administrativeunit ;
     geolinkeddata:nombre "Lugo" ;
     :nationallevelname "Provincia" ;
     geo:hasGeometry ?gs .
  ?t a geolinkeddata:Humedal ;
     geo:hasGeometry ?gt .
  FILTER (geof:sfWithin(?gw, ?gs))
  FILTER (geof:sfWithin(?gt, ?gs))
  FILTER (geof:sfDistance(?gw, ?gt) < 2000)
}

```

https://wmts.mapama.gob.es/sig-api/ogc/features/v1/
collections/agua:Zif_riesgo_pob_q500/items
?bbox={BBOX(?gs)}

?bbox= -8.066664,42.283605,-6.738614,43.807291

agua:zif_riesgo_pob_q500:64084 a :agua:zif_riesgo_pob_q500 .
 agua:zif_riesgo_pob_q500:64084 hasGeometry "POLYGON(...)" .
 agua:zif_riesgo_pob_q500:64217 a :agua:zif_riesgo_pob_q500 .
 agua:zif_riesgo_pob_q500:64217 hasGeometry "POLYGON(...)" .
 agua:zif_riesgo_pob_q500:64646 a :agua:zif_riesgo_pob_q500 .
 agua:zif_riesgo_pob_q500:64646 hasGeometry "POLYGON(...)" .
 ...

?s → administrativenunit:Lugo
 ?gs → "POLYGON(...)"
 ?w → agua:zif_riesgo_pob_q500:64084

Query Evaluation

```

SELECT ?w ?gw WHERE {
  ?w a :agua:Zif_riesgo_pob_q500 ;
    geo:hasGeometry ?gw .
  ?s a :administrativeunit ;
    geolinkeddata:nombre "Lugo" ;
    :nationallevelname "Provincia" ;
    geo:hasGeometry ?gs .
  ?t a geolinkeddata:Humedal ;
    geo:hasGeometry ?gt .
  FILTER (geof:sfWithin(?gw, ?gs))
  FILTER (geof:sfWithin(?gt, ?gs))
  FILTER (geof:sfDistance(?gw, ?gt) < 2000)
}

```

https://wmts.mapama.gob.es/sig-api/ogc/features/v1/
collections/agua:Zif_riesgo_pob_q500/items
?bbox={BBOX(?gs)}

?bbox= -8.066664,42.283605,-6.738614,43.807291

agua:zif_riesgo_pob_q500:64084 a :agua:zif_riesgo_pob_q500 .
 agua:zif_riesgo_pob_q500:64084 hasGeometry "POLYGON(...)" .
 agua:zif_riesgo_pob_q500:64217 a :agua:zif_riesgo_pob_q500 .
 agua:zif_riesgo_pob_q500:64217 hasGeometry "POLYGON(...)" .
 agua:zif_riesgo_pob_q500:64646 a :agua:zif_riesgo_pob_q500 .
 agua:zif_riesgo_pob_q500:64646 hasGeometry "POLYGON(...)" .
 ...

?s → administrativenunit:Lugo
 ?gs → "POLYGON(...)"
 ?w → agua:zif_riesgo_pob_q500:64084

?gw → "POLYGON(...)"

Query Evaluation

```

SELECT ?w ?gw WHERE {
  ?w a :agua:Zif_riesgo_pob_q500 ;
    geo:hasGeometry ?gw .
  ?s a :administrativeunit ;
    geolinkeddata:nombre "Lugo" ;
    :nationallevelname "Provincia" ;
    geo:hasGeometry ?gs .
  ?t a geolinkeddata:Humedal ;
    geo:hasGeometry ?gt .
  FILTER (geof:sfWithin(?gw, ?gs))
  FILTER (geof:sfWithin(?gt, ?gs))
  FILTER (geof:sfDistance(?gw, ?gt) < 2000)
}

```

https://api-features.idee.es/collections/wetland/items
?bbox={BBOX(?gs)}

?bbox= -8.066664,42.283605,-6.738614,43.807291



```

wetland:ES014HYWLDS00000001553510 a geolinkeddata:Humedal .
wetland:ES014HYWLDS00000001553510 hasGeometry "POLYGON(...)" .

wetland:ES014HYWLDS00000001553511 a geolinkeddata:Humedal .
wetland:ES014HYWLDS00000001553511 hasGeometry "POLYGON(...)" .

wetland:ES014HYWLDS00000001553979 a geolinkeddata:Humedal .
wetland:ES014HYWLDS00000001553979 hasGeometry "POLYGON(...)" .

...

```

?s → administrativeunit:Lugo
 ?gs → "POLYGON(...)"
 ?w → agua:zif_riesgo_pob_q500:64084

?gw → "POLYGON(...)"
 ?t → wetland:ES014HYWLDS00000001553510

Query Evaluation

```

SELECT ?w ?gw WHERE {
  ?w a :agua:Zif_riesgo_pob_q500 ;
    geo:hasGeometry ?gw .
  ?s a :administrativeunit ;
    geolinkeddata:nombre "Lugo" ;
    :nationallevelname "Provincia" ;
    geo:hasGeometry ?gs .
  ?t a geolinkeddata:Humedal ;
    geo:hasGeometry ?gt .
  FILTER (geof:sfWithin(?gw, ?gs))
  FILTER (geof:sfWithin(?gt, ?gs))
  FILTER (geof:sfDistance(?gw, ?gt) < 2000)
}

```

https://api-features.idee.es/collections/wetland/items
?bbox={BBOX(?gs)}

?bbox= -8.066664,42.283605,-6.738614,43.807291



```

wetland:ES014HYWLD50000001553510 a geolinkeddata:Humedal .
wetland:ES014HYWLD50000001553510 hasGeometry "POLYGON(...)" .

wetland:ES014HYWLD50000001553511 a geolinkeddata:Humedal .
wetland:ES014HYWLD50000001553511 hasGeometry "POLYGON(...)" .

wetland:ES014HYWLD50000001553979 a geolinkeddata:Humedal .
wetland:ES014HYWLD50000001553979 hasGeometry "POLYGON(...)" .
...

```

?s → administrativenunit:Lugo
 ?gs → "POLYGON(...)"
 ?w → agua:zif_riesgo_pob_q500:64084

?gw → "POLYGON(...)"
 ?t → wetland:ES014HYWLD50000001553510
 ?gt → "POLYGON(...)"

Query Evaluation

```

SELECT ?w ?gw WHERE {
  ?w a :agua:Zif_riesgo_pob_q500 ;
    geo:hasGeometry ?gw .
  ?s a :administrativeunit ;
    geolinkeddata:nombre "Lugo" ;
    :nationallevelname "Provincia" ;
    geo:hasGeometry ?gs .
  ?t a geolinkeddata:Humedal ;
    geo:hasGeometry ?gt .
  FILTER (geof:sfWithin(?gw, ?gs))
  FILTER (geof:sfWithin(?gt, ?gs))
  FILTER (geof:sfDistance(?gw, ?gt) < 2000)
}

```

```

?s -> administrativenunit:Lugo
?gs -> "POLYGON(...)"
?w -> agua:zif_riesgo_pob_q500:64084
?gw -> "POLYGON(...)"
?t -> wetland:ES014HYWLDS0000001553510
?gt -> "POLYGON(...)"

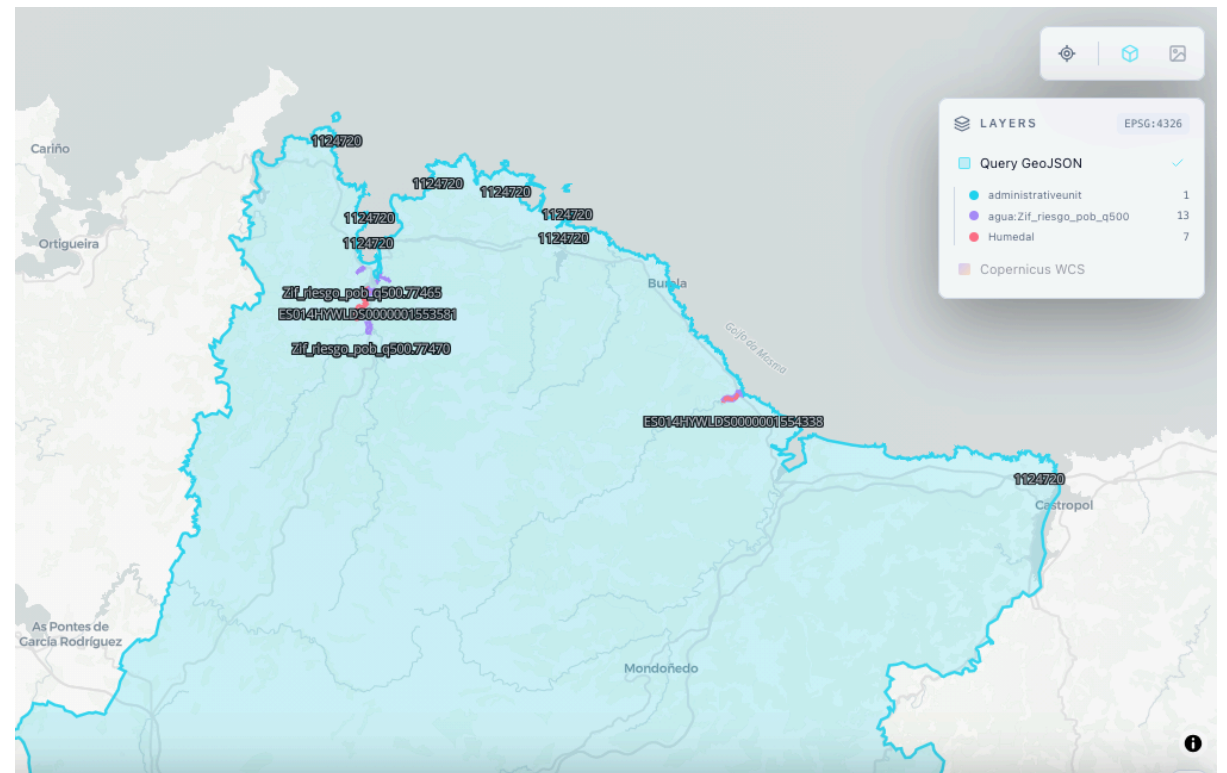
```

If the solution satisfies FILTER constraints, it will be valid

Query Evaluation

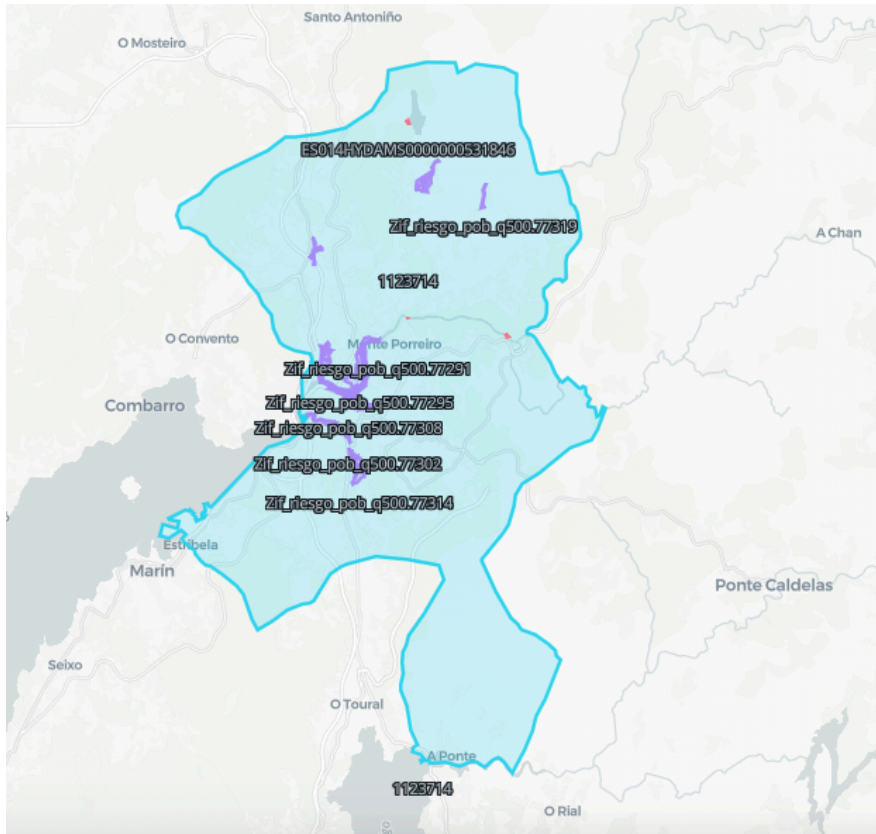
Evaluation will loop, but now
no materialization is needed
as everything is already
materialized

```
SELECT ?w ?gw WHERE {
    ?w a :agua:Zif_riesgo_pob_q500 ;
        geo:hasGeometry ?gw .
    ?s a :administrativeunit ;
        geolinkeddata:nombre "Lugo" ;
        :nationallevelname "Provincia" ;
        geo:hasGeometry ?gs .
    ?t a geolinkeddata:Humedal ;
        geo:hasGeometry ?gt .
    FILTER (geof:sfWithin(?gw, ?gs))
    FILTER (geof:sfWithin(?gt, ?gs))
    FILTER (geof:sfDistance(?gw, ?gt) < 2000)
}
```

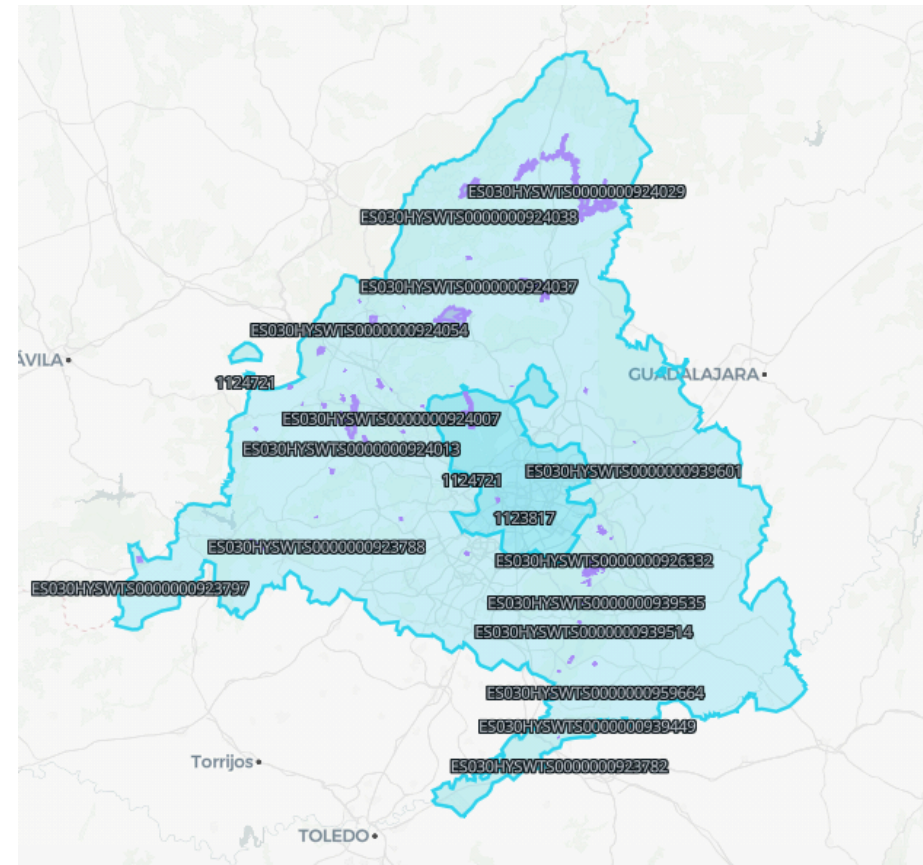


Other Queries

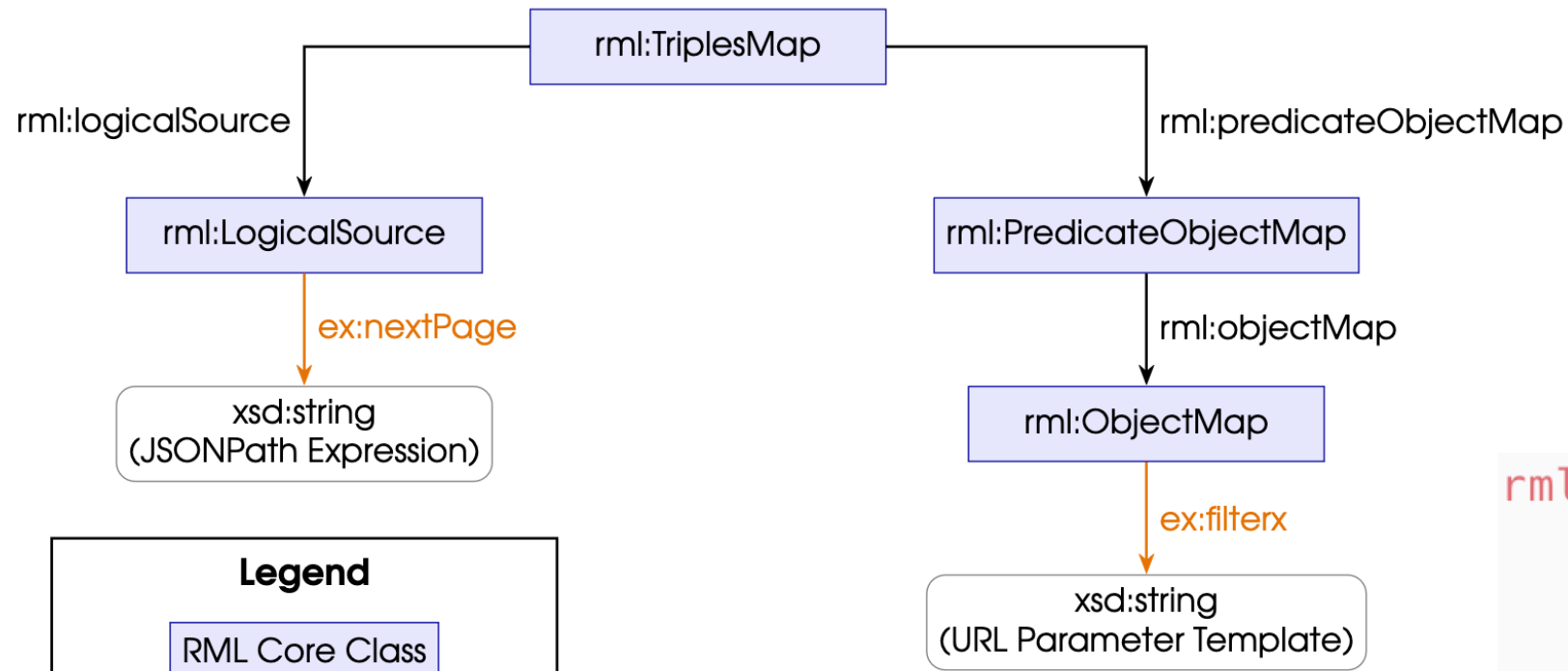
Municipal spatial join over Pontevedra, flood-risk areas, and dams.



Spatial containment over Madrid and standing-water bodies



RML extension



Legend

RML Core Class

Datatype / Literal

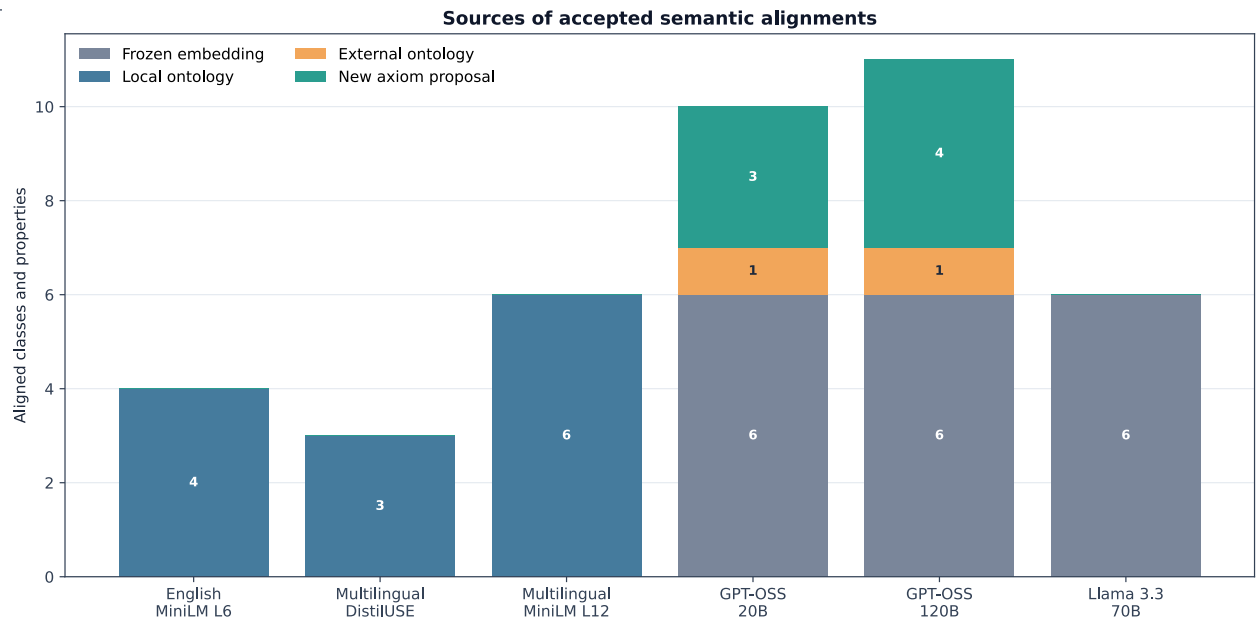
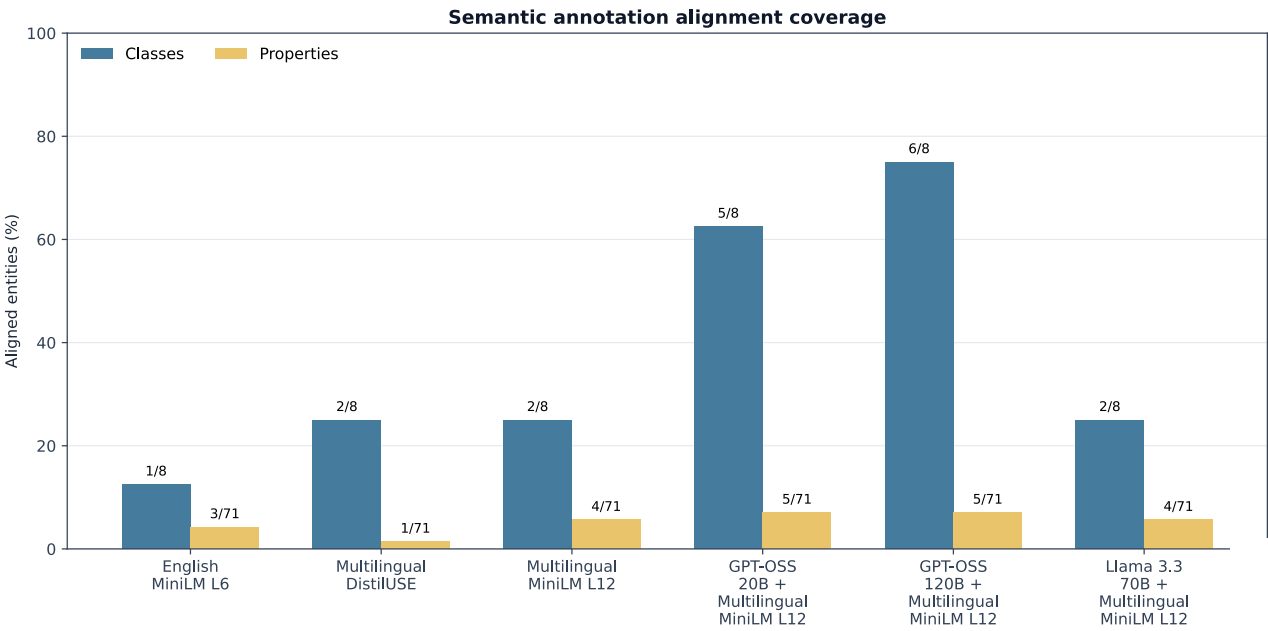
→ Extension Property

Custom vocabulary extension
to augment standard RML
Triples Maps

```

rml:predicate geo:hasGeometry ;
rml:objectMap [
rml:reference "geometry" ;
ex:filterx "bbox=@{1}"
]
  
```

Semantic Annotation Evaluation



VKG Evaluation

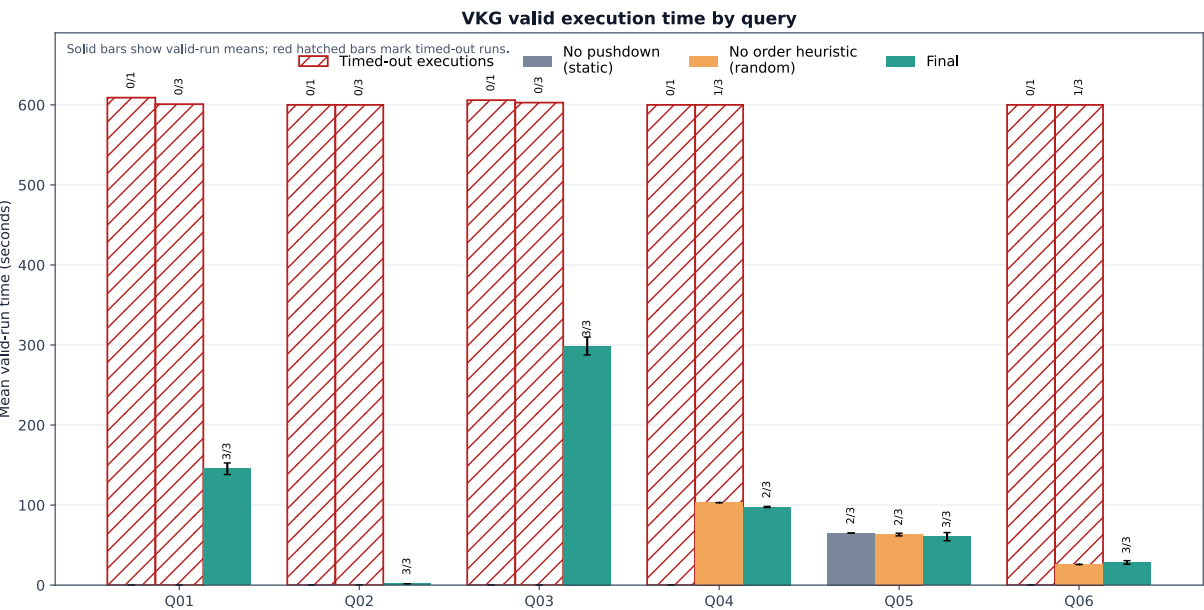


Table 5.4: Observed valid-output profile and mean elapsed time. The *valid* columns report valid executions over executed repetitions; dashes indicate that no valid execution was observed.

Query	Base valid	Base s	Rand. valid	Rand. s	Final valid	Final s
Q01	0/1	—	0/3	—	3/3	145.4
Q02	0/1	—	0/3	—	3/3	1.7
Q03	0/1	—	0/3	—	3/3	298.8
Q04	0/1	—	1/3	103.1	2/3	97.6
Q05	2/3	65.1	2/3	63.2	3/3	60.5
Q06	0/1	—	1/3	25.8	3/3	28.4

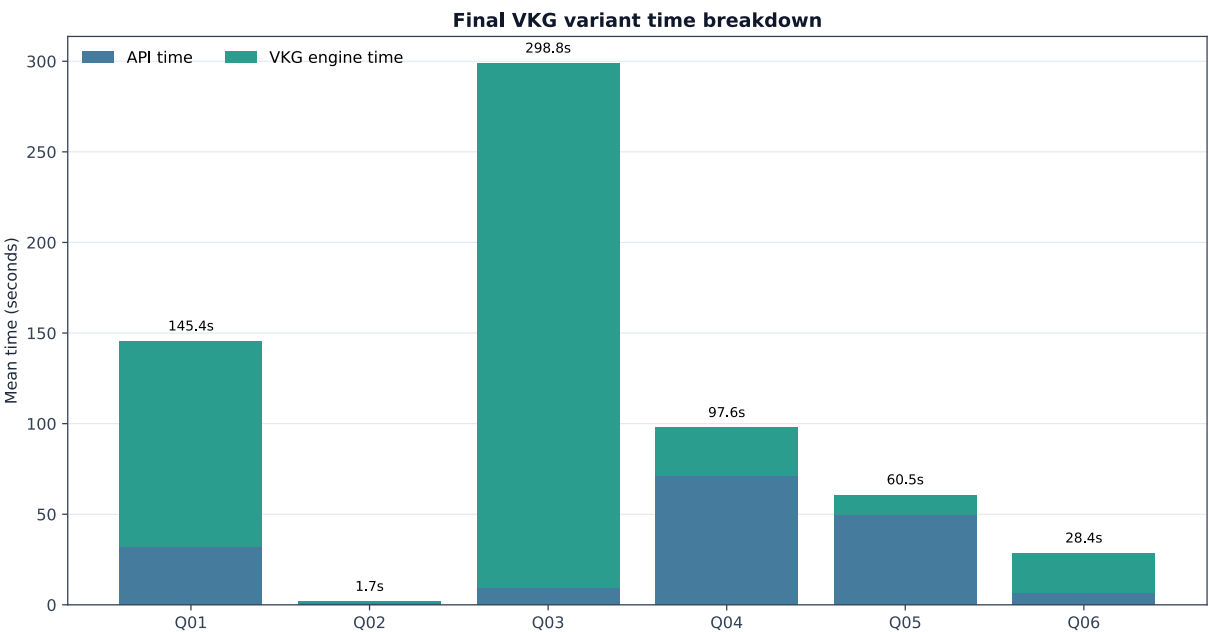


Table 5.3: Execution profile of the final VKG variant over valid repetitions. Response sizes are reported in MB.

Query	Total	API	Engine	Calls	MB	Triples	Rows
Q01	145.4	31.8	113.6	2.0	228.9	2,537	1,181
Q02	1.7	1.0	0.7	3.0	2.6	118	98
Q03	298.8	9.4	289.3	3.0	50.8	802	26
Q04	97.6	71.1	26.6	4.0	268.6	263,472	1
Q05	60.5	49.9	10.6	2.0	118.3	263,424	1
Q06	28.4	6.6	21.7	3.0	40.2	6,903	1,385

Conclusions & Contributions

- **Federated VKG Architecture:** SPARQL/GeoSPARQL querying of JSON Web APIs without full prior materialization.
- **Formal Framework:** Query-specific mapping selection model of the mappings that are sufficient to preserve query answers.
- **RML Extensions:** Incorporation of *filterx* and *nextPage* descriptors for API filtering and pagination.
- **Mapping Generation Pipeline:** Automated generation (OGC API - Features) and human-in-the-loop semantic enrichment.

Future work

- **Triple pattern ordering** via cost-aware optimizer
- Improved **local geospatial execution**
- **OpenAPI/JSON Schema support** in mapping generation
- API Semantic annotation **benchmark/gold standard**

All resources **openly available** in Github¹ and Zenodo²

¹<https://github.com/isaacnoya/FvKG-json>

²<https://doi.org/10.5281/zenodo.20961134>



FvKG[json]: Federated Virtual Knowledge Graph over JSON-based Web APIs

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